What Your Agent Topology Is Actually Doing: Controlled Experiments on Communication Structure in Multi-Agent Systems

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Abstract

Multi-agent AI systems choose communication topologies—star, chain, ring, fully connected—with no controlled evidence for how topology affects behavior. We provide three lines of evidence. First, using island-model genetic algorithms as a model system (240 runs), we show topology independently explains 17% of behavioral diversity variance ($p < 10^{-7}$), amplified 14× on hard problems. Second, we validate the transfer to real agents: 3,200 LLM API calls across 8 coding tasks show that topology modulates solution diversity in GPT-4o-mini agents, with the effect sign determined by the coupling instruction: negative coupling ("use a different algorithm") inverts the GA diversity ordering (Cohen's d from -14.0 to -1.9 across six tasks); positive coupling ("build on and improve") restores it ($\eta^2 = 0.291$, $p < 10^{-19}$). A single instruction change flips which topology produces the most diverse output. The sign depends on whether coupling merges neighbors' solutions (homogenizing, as in GA migration) or excludes them (diversifying, as in the "be different" instruction). At least nine deployed systems chose topologies without controlled comparisons. Topology is a first-class safety variable; current systems are tuning it blindly.

1 Introduction

Every multi-agent system embeds a communication topology—round-robin group chat (AutoGen), DAG pipelines (LangGraph), manager hubs (CrewAI)—yet these choices are made without controlled evidence. Dochkina [Dochkina, 2026] ran 25,000 tasks across 256 agents, finding a 44% quality spread ($d = 1.86$) between topologies. Topology is doing something—but *what, exactly?*

We provide the first controlled answer, bridging from a model system (genetic algorithms) to real LLM agents. Our contributions: (1) topology independently determines behavioral diversity in both GAs and LLM agents; (2) the effect *inverts* between GAs and agents, explained by whether the coupling mechanism merges or constrains; (3) at least nine deployed systems chose topologies without controlled comparisons.

2 Island Models as Agent Systems

An island-model GA partitions a population across sub-populations (*islands*) that evolve independently, exchanging individuals periodically via *migration* channels. The mapping to agent orchestration is direct at the graph level:

GA concept	Agent equivalent
Island	Agent (local state, computation)
Migration channel	Message channel
Population diversity	Solution diversity
Fitness function	Task performance metric
Migration topology	Orchestration graph

Both systems are nodes with local state exchanging information along directed channels. The analogy is weaker at the node level—a GA island contains a *population* evolving over many generations, while an agent produces a single output—but the diversity–topology relationship, our object of study, is a graph-level property in both cases. The agent-side quantities are measured directly: **agent diversity** = mean pairwise cosine distance on sentence embeddings (all-MiniLM-L6-v2) of agent outputs; **agent fitness** = task performance (test pass rate, correctness). GAs provide cheap experimental control (hundreds of trials); the LLM experiment (Section 4.3) validates transfer to real agents.

3 Definitions

Definition 1 (Communication topology)**.** *A* communication topology *for n agents is a directed graph*

$D = (V, A)$ with $|V| = n$, where each arc $(u, v) \in A$ is a permitted channel from agent u to agent v.

Definition 2 (Cycle rank)**.** *The* cycle rank $\beta_1(G) = |E| - |V| + 1$ *(for connected G) counts independent cycles—feedback loops in the communication structure.*

Definition 3 (Directed cycle count)**.** *$\kappa(D)$ counts distinct simple directed cycles in D [Johnson, 1975]. Unlike β_1, this depends on arc* arrangement*, not just edge count. A DAG has $\kappa = 0$.*

NK landscapes [Kauffman, 1993] are tunably rugged fitness functions over binary strings of length N, where each bit's contribution depends on K other bits. K=0 gives a smooth landscape; increasing K adds local optima.

4 Experimental Evidence

We present three experiments spanning GA model systems and LLM agents, totaling 490 GA runs and 3,200 LLM API calls.

4.1 Experiment 1: Topology affects diversity at constant density

Comparing a chain to a complete graph conflates density and cycle structure: $\beta_1 = m - n + 1$ [Hatcher, 2002] means cycle count and edge count are perfectly correlated for undirected graphs. We escape via directed cycle count κ, which varies independently of density.

Design. 8 directed graphs, all n=8, 16 arcs (δ=2.0), varying only in $\kappa \in \{0, 0, 3, 10, 14, 20, 29, 47\}$. Island-model GA: 50 individuals/island, 500 generations, 30 seeds per topology, 240 runs total.

Results. $F(7, 232) = 6.90$, $p = 1.85 \times 10^{-7}$, $\eta^2 = 0.17$. More directed cycles *accelerate consensus* ($r = -0.68$): topologies with more feedback loops cause agents to converge faster, reducing capacity to maintain alternative strategies.

4.2 Experiment 2: Hard problems amplify the effect 14×

On NK landscapes with tunable ruggedness (250 runs: $K \in \{0, 2, 4, 6\}$, topologies selected to span the κ range from Experiment 1, 10–20 seeds per condition), topology's effect scales with difficulty:

K	η^2 (diversity)	Interpretation
0 (trivial)	0.05	Topology barely matters
2 (moderate)	0.20	Moderate effect
4 (hard)	0.45	Large effect
6 (very hard)	0.69	Topology dominates

Simple tasks are structurally smooth—topology barely matters. Complex tasks (multiple valid strategies, ambiguous evaluation) are rugged—topology dominates. Real-world deployments overwhelmingly involve the latter.

4.3 Experiment 3: Topology inverts diversity in LLM agents

The GA experiments establish that topology controls diversity. But does this transfer to LLM agents? We test directly.

Design. 5 GPT-4o-mini agents, 8 coding tasks (shortest path, kth largest, rate limiter, JSON parser, LRU cache, regex matcher, expression evaluator, Bloom filter), 4 topologies (none, ring, star, fully connected), 10 seeds per condition. 1,600 API calls total. Coupling is *negative*: each agent is instructed to use a *different* algorithm from those visible in its context. Diversity is measured as mean pairwise cosine distance on sentence embeddings (all-MiniLM-L6-v2) of agent outputs.

Metric validation. The "none" topology provides an implicit sanity check: all five agents generate independently with no information sharing, yet produce near-identical outputs (mean diversity 0.090 across tasks, $\sigma \approx 0.01$)—confirming the metric detects the known homogeneity of the model's prior distribution. Under connected topologies with negative coupling, where agents are explicitly instructed to use different algorithms, diversity rises substantially. The low within-condition variance also explains the large Cohen's d values reported below: the effect is clean, not an artifact of the scale.

Results. The diversity ordering **inverts** from the GA result:

System	**Diversity ordering**	**Statistic**
GA	none > ring > FC	$W = 1.0$ (6 domains)
LLM	FC > ring ≈ star > none	$W = 0.91$ (6/8 tasks)

Six of eight tasks show the inverted ordering cleanly (Kruskal-Wallis $p < 10^{-6}$; Cohen's d ranges from

-14.0 to -1.9). Two exceptions are informative: Bloom filter ($d \approx 0$, a unimodal task with essentially one canonical implementation—no algorithmic variety for topology to act on) and expression evaluator ($d = +0.4$, positive: the model's prior is already spread across multiple parsing strategies). The expression evaluator confirms the mechanism's boundary condition: when the prior is already flat (high entropy), isolation produces diverse solutions—the GA regime—and negative coupling adds little. The sign flip requires a peaked prior.

Why the sign flips. Under "none" topology, all 5 LLM agents independently generate the *same* algorithm (e.g., quickselect for kth_largest). LLMs start from a **peaked prior** (low entropy); GAs start from **random initialization** (high entropy). Isolation preserves initial entropy: high for GAs (diverse), low for LLMs (homogeneous). The inversion follows from the coupling direction:

- **GA migration** is colimit-like: it merges populations by injecting solutions into neighboring islands. More connectivity → faster homogenization → less diversity.
- **LLM "be different" context** is limit-like: it accumulates exclusion constraints from neighbors. More connectivity → more constraints → agents forced into distributional tails → *more* diversity.

The topology is the invariant; the sign depends on the coupling direction. We conjecture that this sign reversal is an instance of the adjunction between left and right Kan extensions along the topology functor—colimit (merge) vs. limit (constrain)—with the formal categorical construction developed in a companion paper [**?**].

Positive coupling confirms the prediction. We repeated the experiment with *positive* coupling: each agent is instructed to "build on and improve" neighbors' solutions instead of "use a different algorithm." 1,600 additional API calls (total: 3,200). Results: the diversity ordering **reverts to GA-like**: none > star ≈ ring > FC (Kruskal-Wallis $H = 94.83$, $p = 2.01 \times 10^{-20}$; Cohen's $d = 1.08$, none vs. FC; $\eta^2 = 0.291$). Crucially, the *none* topology—where agents have no neighbors—shows identical diversity across conditions (negative: 0.090, positive: 0.095, $p = 0.96$), confirming that coupling direction, not some other experimental variable, drives the reversal. Connected topologies collapse to near-zero diversity (0.025–0.028) under positive coupling, as agents converge on a shared best solution. A single instruction change flips the diversity ordering—precisely as the colimit/limit framework predicts.

Table 1: Agent systems whose topology choices lack controlled evidence.

System	Issue
Graph-GRPO [Cang et al., 2026]	Complete = worst; also max β_1
OFA-MAS [Li et al., 2026]	Prunes density & cycles together
G-Designer [Zhang et al., 2025]	No density-controlled baseline
GoAgent [Chen et al., 2026]	DAG vs. dense alternatives
ARG-Designer [Li et al., 2026b]	DAG space only
AgentConductor [Wang et al., 2026]	Cycles never tested
DyTopo [Lu et al., 2026]	DAG enforced throughout
HERA [Li and Ramakrishnan, 2026]	Cycles tracked, not controlled
Dochkina [Dochkina, 2026]	Density & cycles confounded

5 Implications for Agent Orchestration

The evidence gap. At least nine recent agent systems—spanning ICML, AAAI, WWW, and NeurIPS—chose topologies without density-controlled comparisons (Table 1).

Safety risks. Our LLM experiment sharpens three concerns: *(1) The sign matters.* A single instruction change flips which topology produces the most diverse output ($\eta^2 = 0.291$, $p < 10^{-19}$). Predictions based on GA intuition alone will have the wrong sign for diversifying agents. *(2) Premature consensus.* Isolated LLM agents converge to identical solutions (Cohen's d from -14.0 to -1.9 vs. connected agents across six tasks). "Ensemble diversity" claims based on independent agents may be illusory when models share a peaked prior. *(3) Error amplification.* Moran [Moran, 2025] reports 17× error amplification in unstructured communication. The standard fix (impose DAG structure) simultaneously removes cycles *and* density—we cannot know which helps.

Recommendations.

1. **Treat topology as a first-class variable.** It controls the diversity–convergence tradeoff—in both GAs and LLM agents, empirically confirmed.
2. **Know your coupling direction.** A single instruction change flips the diversity ordering. Merging neighbors' outputs homogenizes; differentiating from them diversifies.
3. **Match topology to task complexity.** Simple tasks tolerate any topology; hard tasks demand deliberate choice (η^2 scales 14× from K=0 to K=6).
4. **Monitor diversity, not just accuracy.** Topology effects manifest in strategy diversity. Accu-

racy alone masks topology-induced failures.

Limitations. Our GA experiment (Experiment 1) controls density by using iso-dense directed graphs. Our LLM experiment does not: the four topologies (none, ring, star, fully connected) differ in both density and structure. The *sign-flip* finding is robust to this confound, since both coupling conditions use identical topologies and the reversal is within-topology. However, the absolute diversity ordering across topologies in the LLM experiment is confounded by density—the same critique we level at prior work. Density-controlled LLM experiments (using iso-dense directed topologies) are needed to fully isolate the mechanism. Additionally, our diversity metric (cosine distance on sentence embeddings) is validated only implicitly via the "none" baseline; a systematic comparison against human-judged algorithmic similarity would strengthen the measurement.

6 Related Work

Yang et al. [Yang et al., 2025] identify topology as the underexplored MAS dimension; we provide the controlled experiments they call for. Dochkina [Dochkina, 2026] shows massive effects (25K tasks, $d = 1.86$) without isolating the mechanism. HERA [Li and Ramakrishnan, 2026] tracks cycle count during evolution but does not control density. In evolutionary computation, the topology–diversity link is established [Tomassini, 2005, Skolicki and De Jong, 2005] without density control. Capucci and Myers [Capucci and Myers, 2026] formalize the cost: each cycle requires an independent convergence guarantee. Langer and Turing [**?**] develop the categorical framework (lax monoidal functors, Kleisli morphisms over an evolution monad) that motivates the colimit/limit interpretation of the sign flip; our Experiment 3 provides the first empirical test of their predicted coupling-sign dependence.

7 Conclusion

Communication topology is not neutral infrastructure—it is a design variable controlling how multi-agent systems balance exploration against convergence. In GA model systems, topology independently explains 17% of diversity variance on easy problems, rising to 69% on hard ones. In LLM agents, the effect is equally strong but its *sign* depends on coupling direction: negative coupling ("be different") inverts the GA ordering (d from -14.0 to -1.9); positive coupling ("build on") restores it ($\eta^2 = 0.291$). The sign is determined by whether coupling merges solutions (colimit-like, homogenizing) or constrains them (limit-like, diversifying)—confirmed by the null control: isolated agents show identical diversity regardless of coupling instruction ($p = 0.96$). At least nine deployed systems chose topologies based on comparisons that simultaneously varied the property they claimed to test. For safe multi-agent deployment, topology must be tuned deliberately, its coupling direction understood, and its effects monitored through diversity metrics—not assumed away.

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